

Introduction

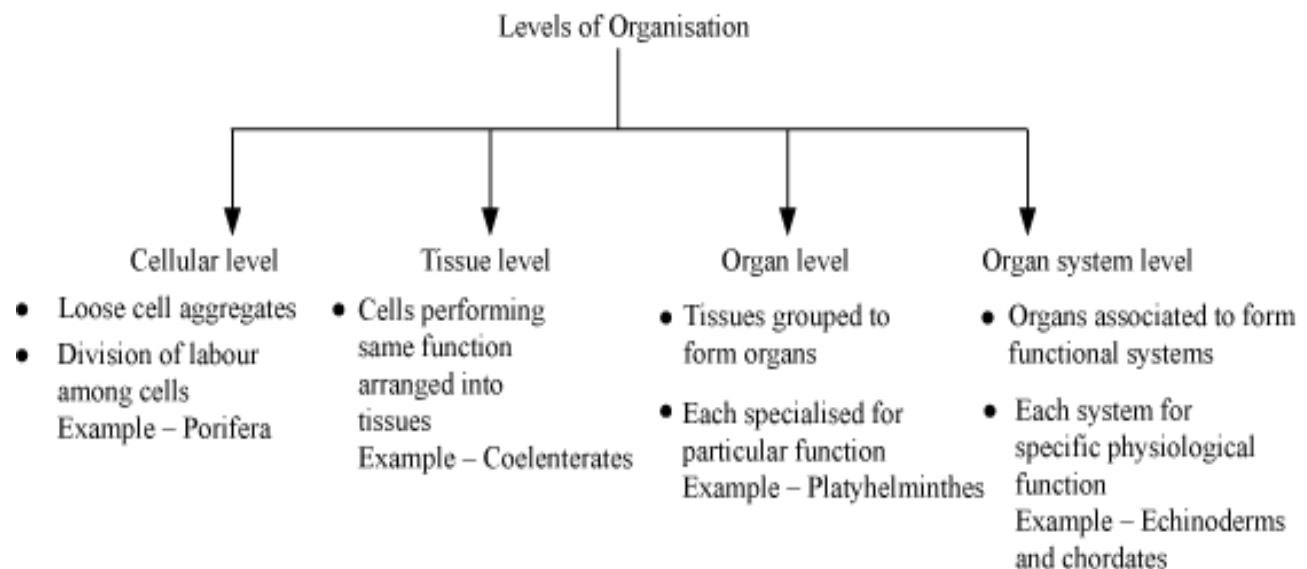
When you look around, you will observe different animals with different structures and forms. As over a million species of animals have been described till now, the need for classification becomes all the more important. The classification also helps in assigning a systematic position to newly described species.

Basis of classification for Kingdom Animalia

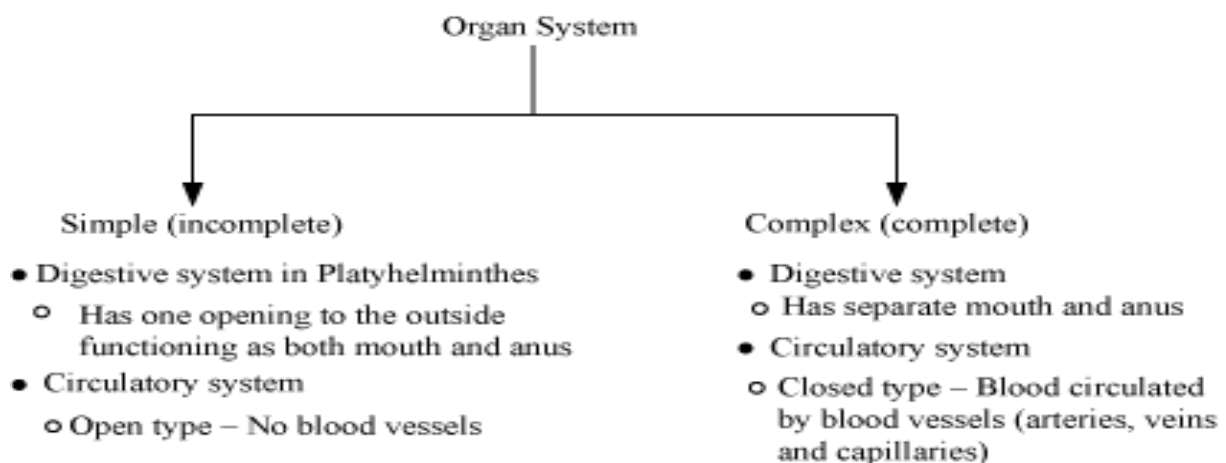
1. levels of organisation (Arrangement of cells)
2. Patterns of digestive, circulatory, or reproductive systems are some of the features used to classify animals.
3. Body symmetry
4. Arrangement of cells in germ layer
5. Nature of coelom
6. Segmentation
7. Presence or absence of notochord

1. Levels of Organisation

- All members of Animalia are multicellular, but exhibit different levels of organisation.

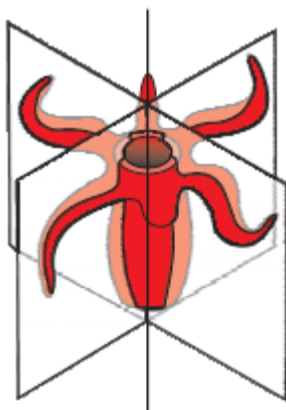
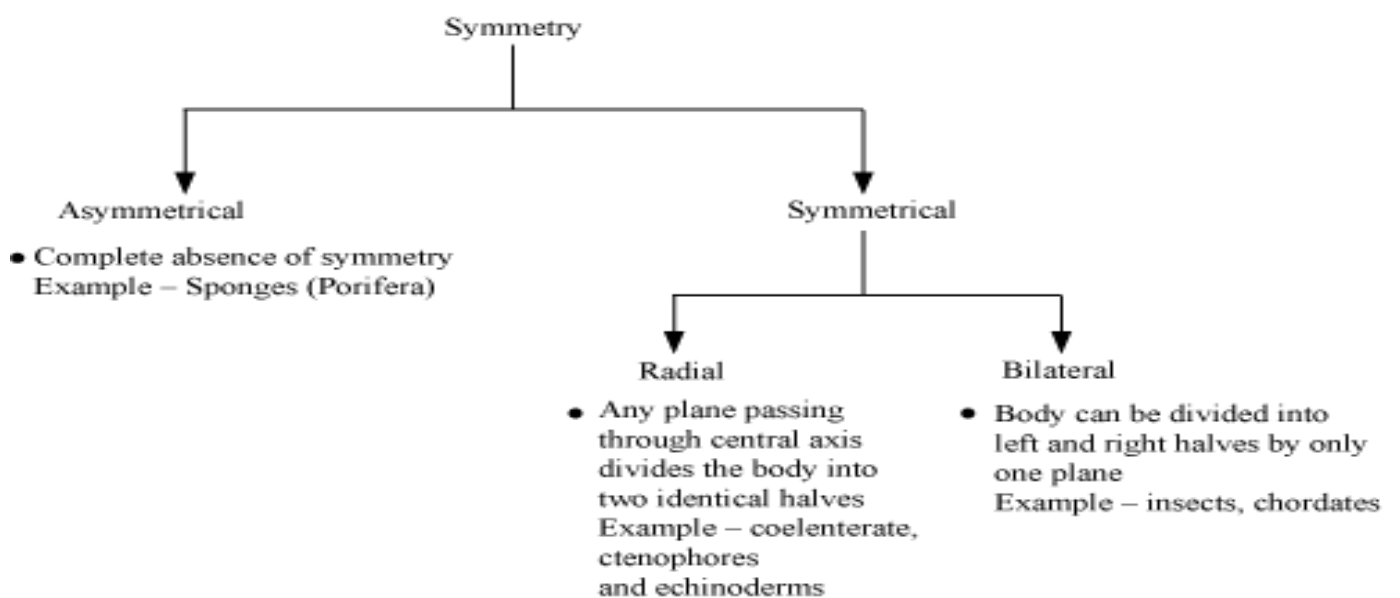


2. Patterns of digestive, circulatory, or reproductive systems

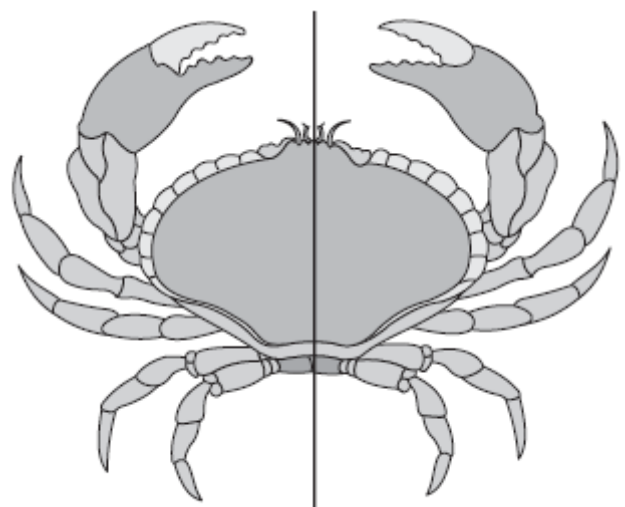


3. Body Symmetry

- Planes passing through the body dividing it into equal halves **symmetry**.
- There are two types of symmetry
 - Radial symmetry:** - Any plane passes through the central axis of the body divides the organism into two identical halves (parts) is called radial symmetry.
 - Bilateral symmetry:** - A plane passing through the central axis of the body dividing the organism into two identical left and a right half is called bilateral symmetry.
- When any plane passes through the centre of the body not dividing the organism into equal halves is called **Asymmetry**.

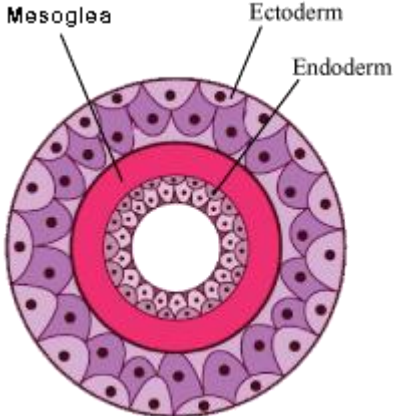
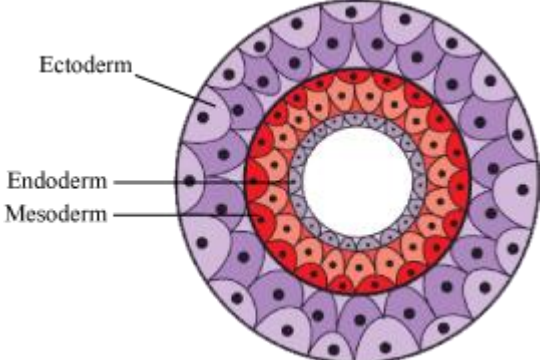


(i) Radial symmetry



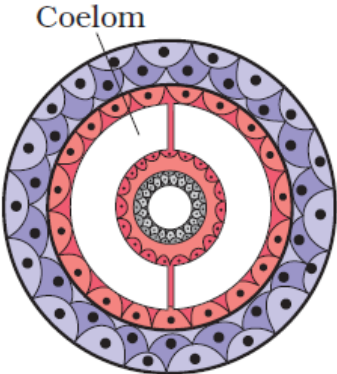
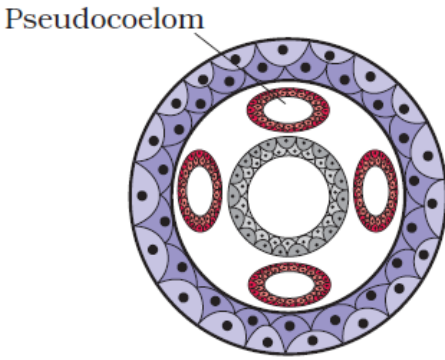
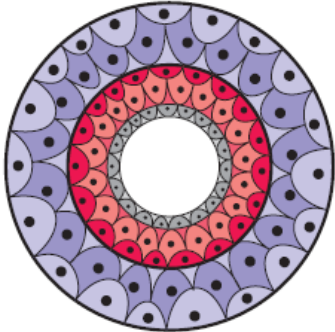
(ii) Bilateral symmetry

4. Arrangement of cells in Embryonic (germ) layer: -

Diploblastic	Triploblastic
- Cell arranged in two embryonic layers: - Ectoderm (external layer) - Endoderm (internal layer) - Presence of an undifferentiated layer, i.e. mesoglea, in between two embryonic layers - Example – coelenterates	- Cells arranged in three embryonic layers: - Ectoderm (external layer) - Mesoderm (middle layer) - Endoderm (internal layer) - Example – Platyhelminthes and chordates
	

5. Nature of Coelom

- Coelom** – Mesoderm-lined body cavity present in between body wall and gut wall is called Coelom.
- Based on the presence or absence of coelom animals are classified into:-

Coelomates	Pseudocoelomates	Acoelomates
- Animals possessing the coelom are called Coelomates. e.g., Annelids, molluscs, arthropods, echinoderms, hemichordates and chordates.	- In some animals, the body cavity is not lined by mesoderm; instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm. Such a body cavity is called pseudocoelom and the animals possessing them are called Pseudocoelomates, e.g., aschelminthes	The animals in which the body cavity is absent are called acoelomates, e.g., Platyhelminthes
		

6. Segmentation

- In some animals, body is externally and internally divided into segments with certain organs repeated serially.
- Example: Earthworm

NOTE: - The body showing segmented pattern called **metameric segmentation** and the phenomenon is known as **metamerism**

7. Presence or absence of Notochord

- Notochord is a mesoderm derived rod-like structure, which is formed dorsally in some animals during embryonic development.
- Animals with notochord are called **chordates**.
- Animals which do not form or possess notochord are called **non-chordates**, e.g., porifera to echinoderms.

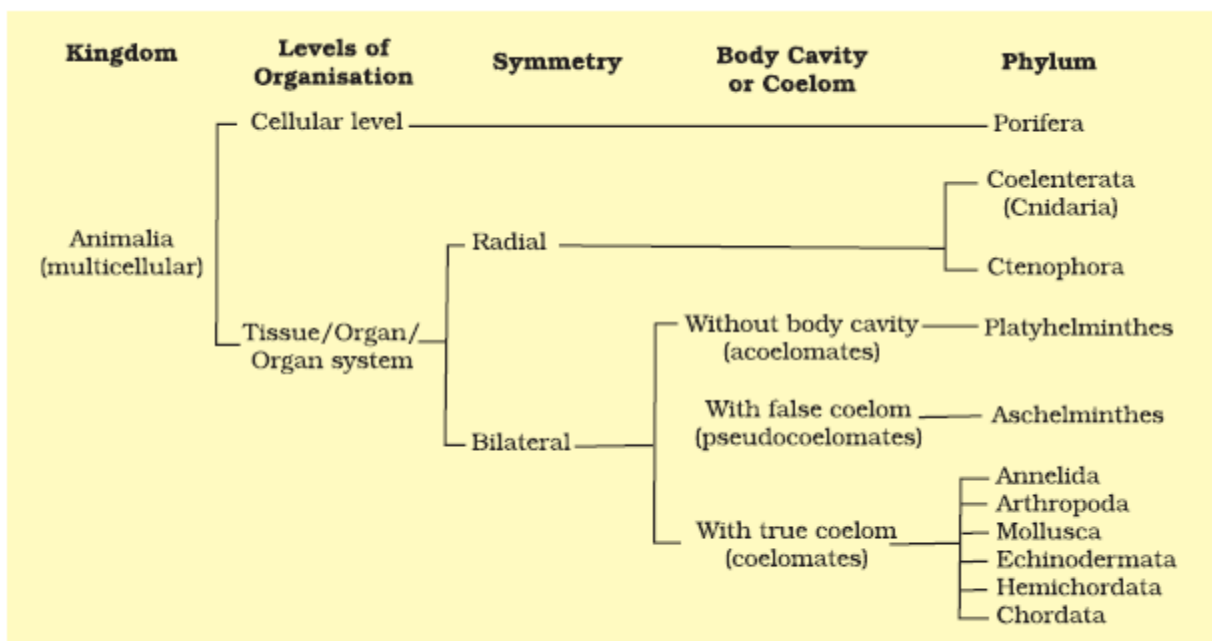
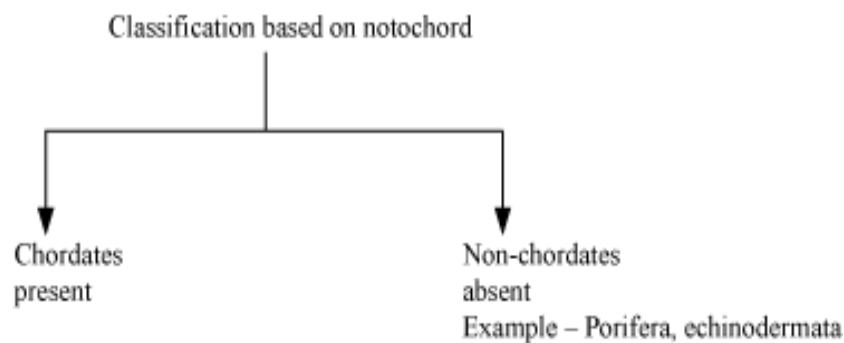
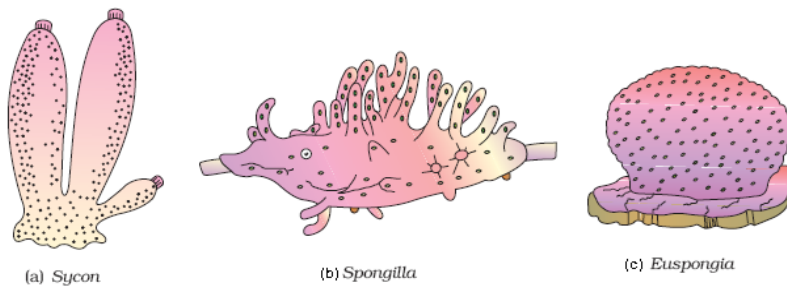


Fig:- Broad classification of Kingdom Animalia based on common fundamental features

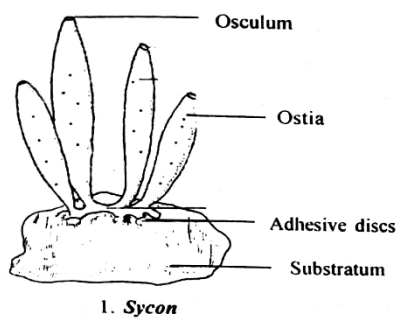
Characteristics Features of Different Phyla

I. Phylum – Porifera (Gr- Porous = pore; ferre = to bear)

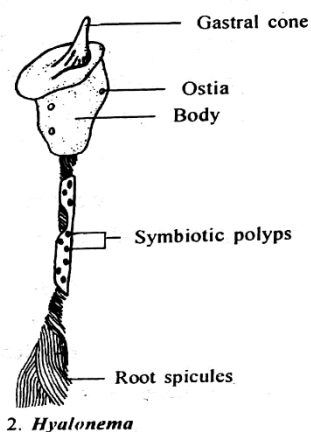
- Members commonly known as sponges are generally marine.
- They are primitive, multicellular, asymmetrical, and have cellular level of organization.
- They have a water transport system or **canal system**.
- Canal system is helpful in gathering food, exchange of respiratory gases, and excretion of waste.
- Waters enters through minute pores (**Ostia**) in the body wall into a central cavity, **Spongocoel** and goes out through Osculum.
- The spongocoel and canals are lined by **collar cells (choanocytes)**.
- Digestion is Intracellular.
- Skeletal system – Skeleton consists of **spicules** or **sponging fibres**.
- Reproduction:
 - Asexual (by fragmentation, budding, gemmules) or sexual (by gamete formation)
 - Sponges are **hermaphrodite/bisexual/monoecious**
 - Fertilization – Internal.
 - Development - Indirect (have morphologically distinct larval stage) (amphiblastula larva in sycon and parenchymula larva in leucosolenia)
- Examples of sponges – (a) **Sycon**, (b) **Spongilla** (fresh water sponge), and (c) **Euspongia** (Bath sponge)



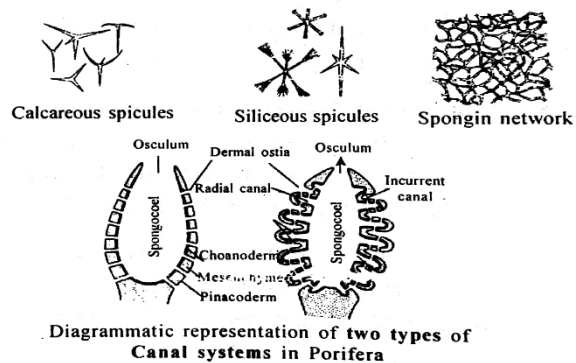
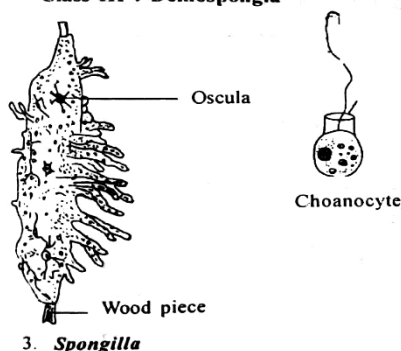
Class I : Calcarea



Class II : Hexactinellida

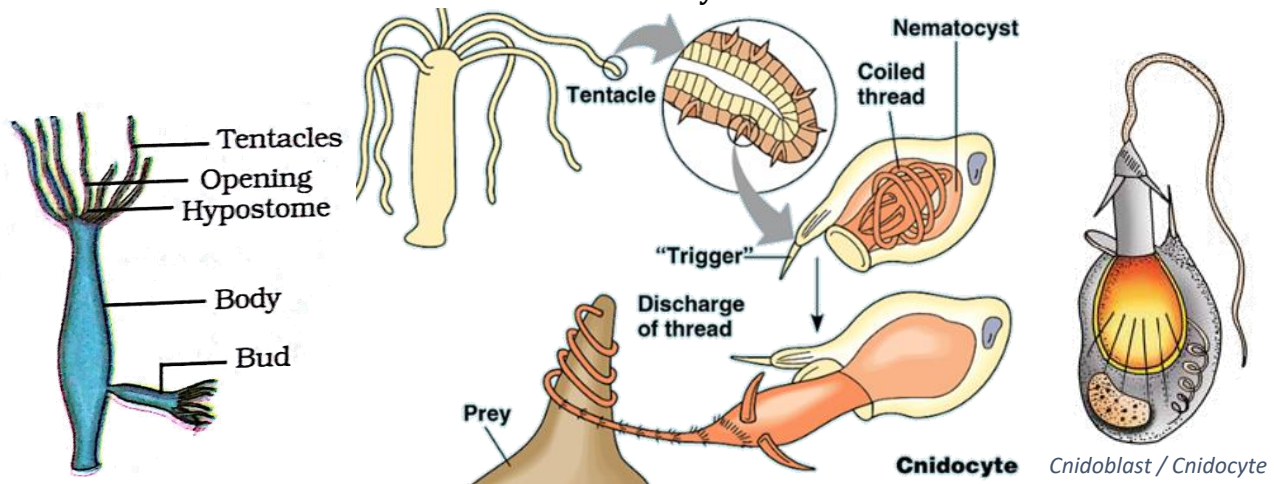


Class III : Demospongia

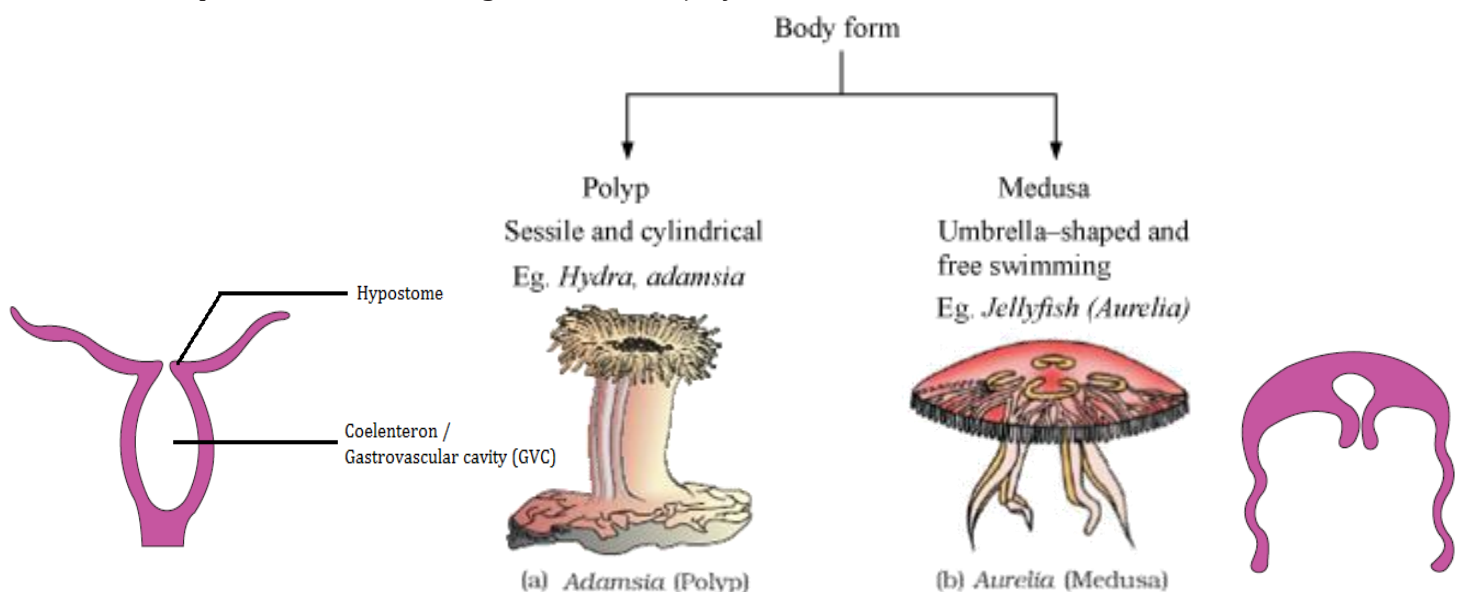


II. Phylum – Coelenterata (Gr. Coel = cavity; enteron = intestine) / Cnidaria (Gr. Knite = nettle; aria = like)

- Coelenterates are aquatic (mostly marine), sessile, or free swimming.
- Radially symmetrical.
- Diploblastic and have tissue level of organization.
- They have a central **gastro-vascular cavity (GVC)** with a single opening, **hypostome**.
- Hypostome is surrounded by tentacles
- Tentacles consists of **cnidoblasts** or **cnidocytes** (which contain the stinging capsules or nematocytes), which are used for anchorage, defense and for capture of prey.
- Name Cnidaria derived from **cnidoblasts** or **cnidocytes**



- Digestion can be internal, extracellular, and intracellular.
- Skeletal system - Skeleton is composed of calcium carbonate (corals).
- Cnidarians exhibit two basic body forms called **polyp** and **medusa**.
- The polyp is a sessile and cylindrical form like *Hydra*, *Adamsia*, etc. whereas, the medusa is umbrella-shaped and free-swimming like *Aurelia* or jelly fish.



- Those cnidarians which exist in both forms exhibit alternation of generation (**Metagenesis**) i.e., polyps produce medusae asexually and medusae form the polyps sexually (e.g., *Obelia*).

{Polyps → medusa (asexually) and medusa → polyps (sexually) (Example – *Obelia*)}

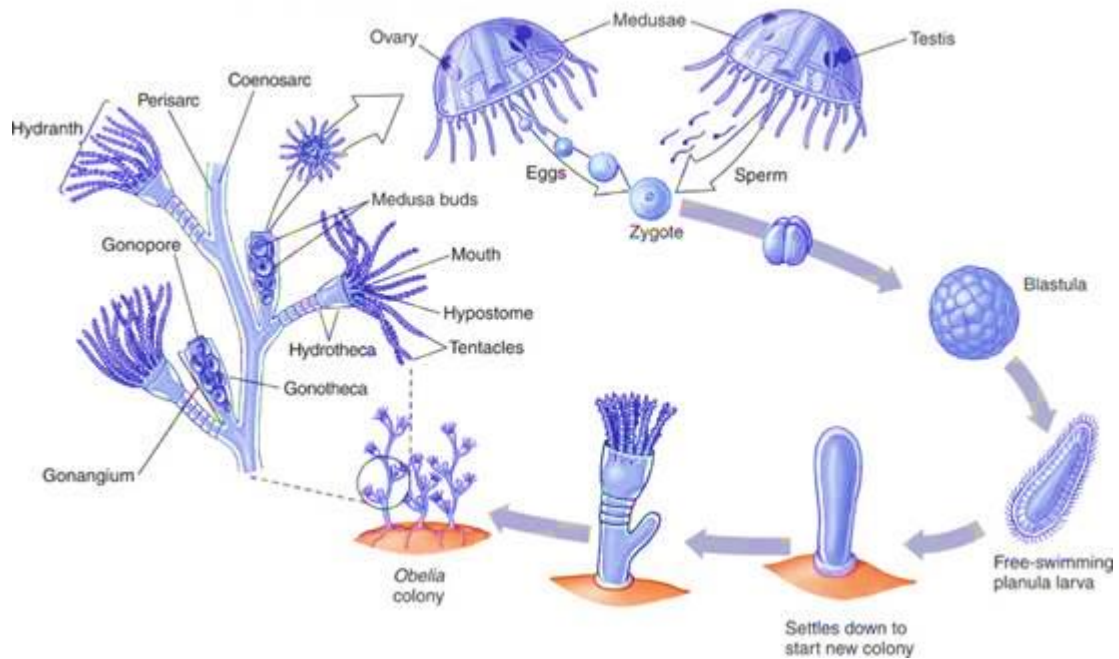


Figure: life cycle of Obelia showing metagenesis

- Examples of coelenterates – *Physalia* (Portuguese man of war), *Adamsia* (Sea anemone), *pennatula* (Sea-pen), *Gorgonia* (Sea-fan), *Meandrina* (Brain coral), *Obelia* (Sea fur), *Aurelia* (jelly fish) etc.



Physalia (Portuguese man of war)



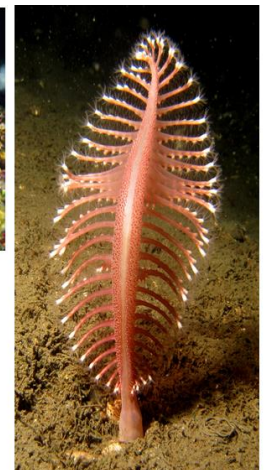
Meandrina (Brain coral)



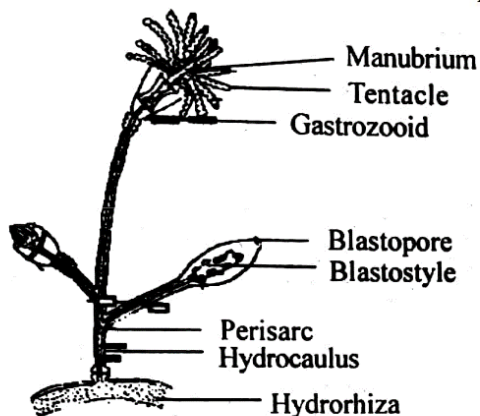
Gorgonia (Sea-fan)



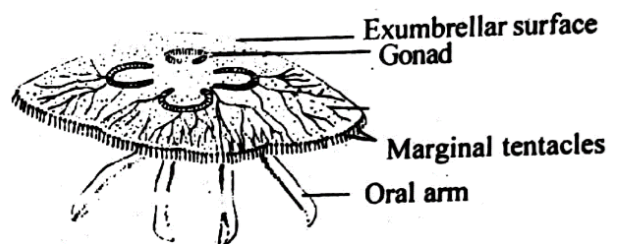
Adamsia (Sea anemone)



Pennatula (Sea-pen)



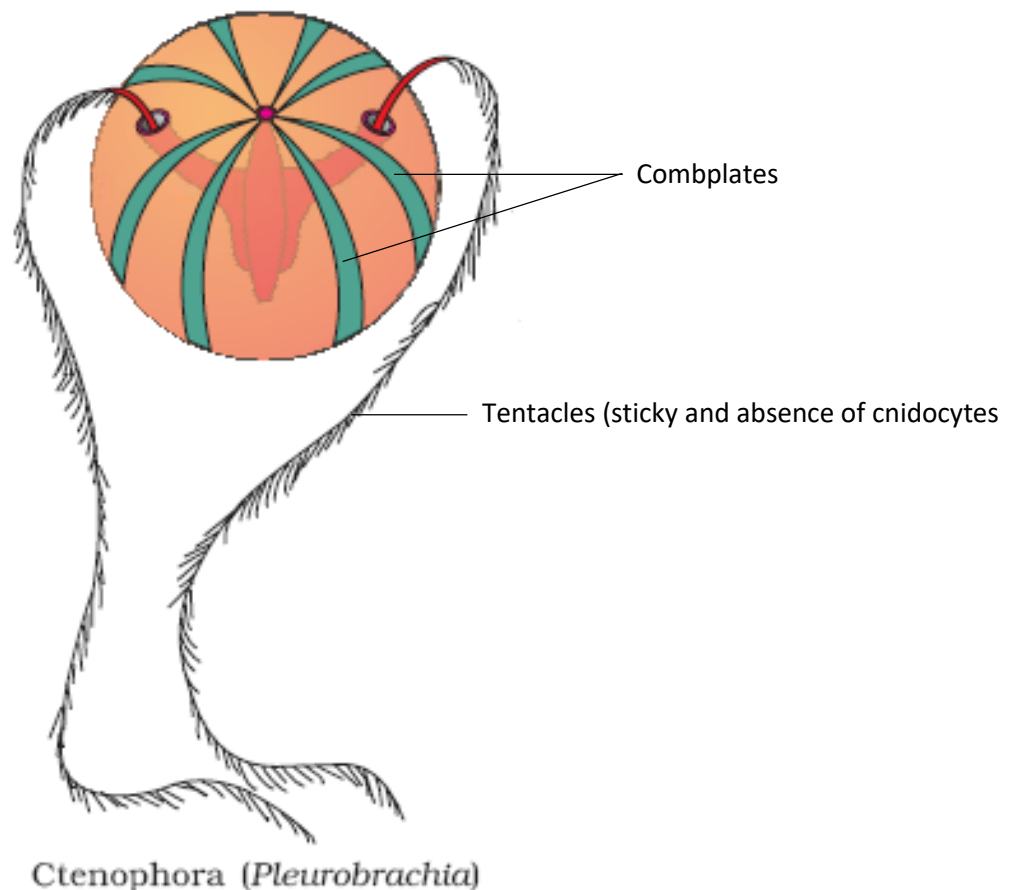
1. *Obelia*



2. *Aurelia*

III. Phylum – Ctenophora (Gr. Ktene = comb; phors = bearing)

- Commonly known as - **Sea walnuts** and **comb jellies** or **sea gooseberry**.
- Exclusively marine.
- Radially symmetrical.
- Diploblastic and have tissue level of organization.
- Digestion - Extracellular or intracellular.
- Locomotory organ - Eight rows of ciliated comb plates located externally.
- Sexes are not separate (hermaphrodite).
- Reproduction - Sexual only
 - Fertilization - external
 - Development – indirect (cydippid larva)
- Exhibit the property of **Bioluminescence** (the property of a living organism to emit light.)
- Examples of ctenophores – ***Pleurobrachia*** (sea gooseberry) and ***Ctenoplana*** (sea walnuts).



IV. Phylum – Platyhelminthes (Gr. Platy = flat; helminthes = worm)

- Platyhelminthes are commonly called as **flatworms** since they have **dorso-ventrally flattened bodies**.
- Members are bilaterally symmetrical.
- Triploblastic and acoelomates with organ level of organisation.
- Nutrition – Parasitic (endoparasites on animals and humans.)
Parasitic forms may have **hooks** and **suckers** present on their bodies or some forms may suck nutrients directly from their host.

- Excretion - Through specialised cell called **Flame cells/Solenocytes/protonephridia** which helps in osmoregulation and excretion.
- Reproduction - Sexually and Asexually (fission).
- Platyhelminthes are **hermaphrodites** (i.e. sexes are not separate.)
- Fertilization - Internal.
- Development - consists of several larval stages.
- Some have high regeneration capacity.

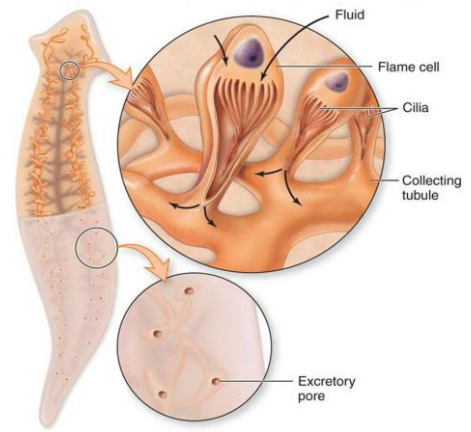


Figure: Excretory system

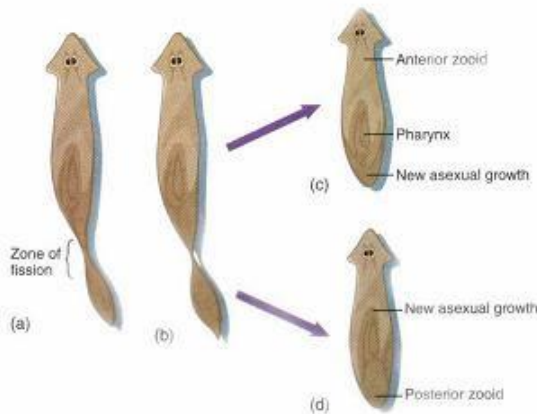


Figure: Asexual reproduction by fission

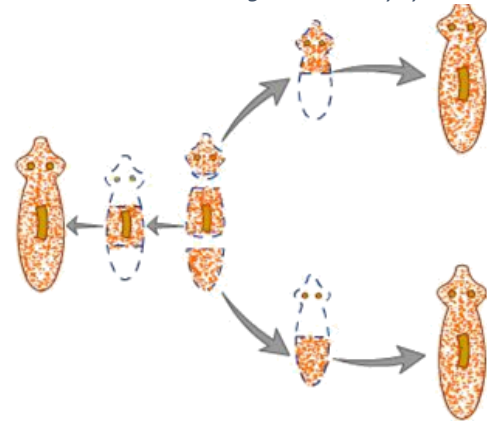
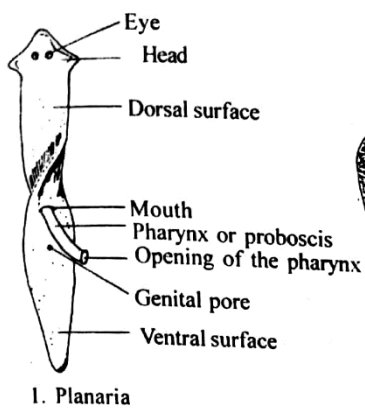


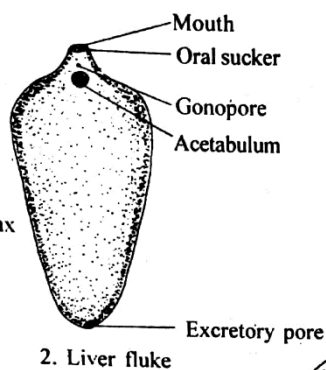
Figure: Planaria showing high regeneration capacity

- Examples of Platyhelminthes – **Taenia** (Tapeworm) (parasite in pigs and in humans) **Fasciola** (Liver fluke)(parasite in goat, sheep), **Dugesia** (Planaria) (aquatic, free-living)

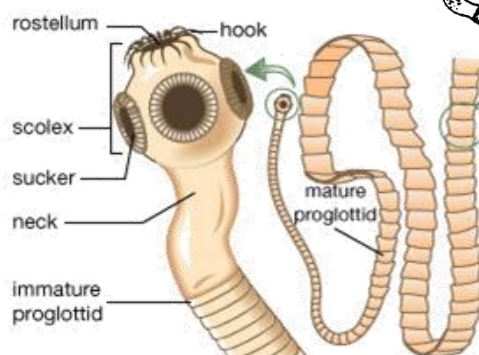
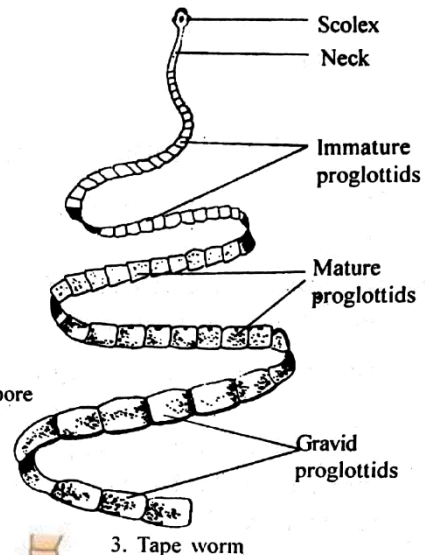
Class I : Turbellaria



Class II : Trematoda



Class III : Cestoda



SYSTEM

V. Phylum –Aschelminthes (Gr. Askos = sac like; helminthes = worm) / Nematyhelminthes (nema = thread; helminthes = worm)

- Known as **round worms** since their body is circular on cross section.
- Have bilateral symmetry.
- Are Triploblastic, Pseudocoelomates, and have organ system level of organisation.
- Nutrition – May be free-living, aquatic and terrestrial or parasitic on plants and animals.
- Alimentary canal is complete with well-developed **muscular pharynx**.
- Excretion – Excretory tube removes waste through the excretory pore.

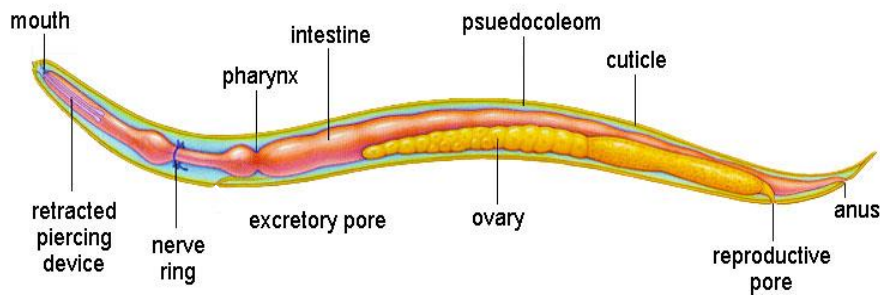


Figure: Diagram showing digestive system and excretory system

- Reproduction – Aschelminthes are dioecious (sexes separate). Females are often longer than males.
 - Fertilization – Internal
 - Development – Direct (young ones resemble adult) or indirect
- Examples of Aschelminthes – *Ascaris* (Round worm), *Wuchereria* (Filarial worm), *Ancylostoma* (Hookworm)

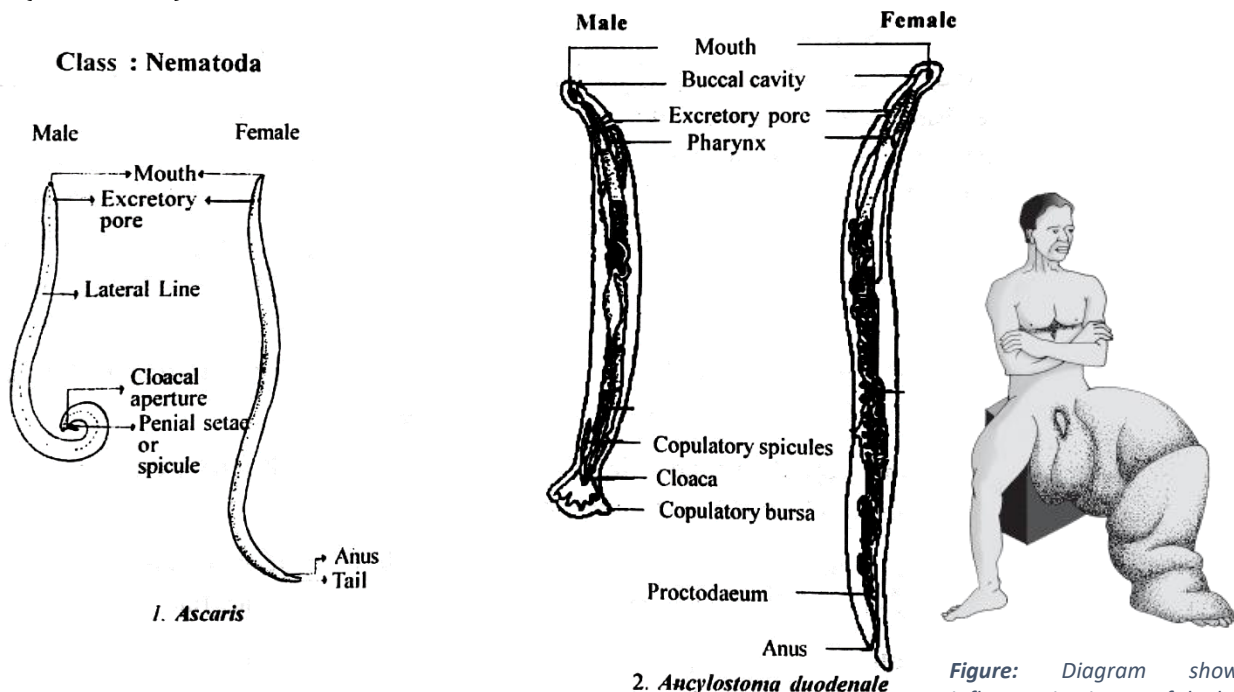


Figure: Diagram showing inflammation in one of the lower limbs due to elephantiasis

NOTE – *Ascaris* causes Ascariasis in human and *Wuchereria* causes Filariasis / Elephantiasis in humans which is transmitted by the wide range of mosquitoes depending upon the geographical area. In Africa the most common vector is *Anopheles* and in America it is *Culex*. *Aedes* and *Mansonia* can transmit the infection in the specific and in Asia. *Aedes* and *Mansonia* also transmits Chikungunya and dengue pathogens. *Ancylostoma* infects small intestine in Cats, Dogs and Puppies

VI. Phylum – Annelida (Latin, Annulus = little ring; eidos = form)

- May be aquatic (marine and freshwater) or terrestrial.
- Members have bilateral symmetry.
- Have organ-system level of organisation, are triploblastic, coelomate, and have **metamerically segmented** bodies.
- Locomotory organs - Longitudinal and circular muscles; **parapodia** in aquatic forms (help in swimming.)
- Circulatory system – Closed.
- Excretory organ – **Nephridia**.
- Neural system - Consists of paired ganglia connected to a double ventral nerve cord by lateral nerves.

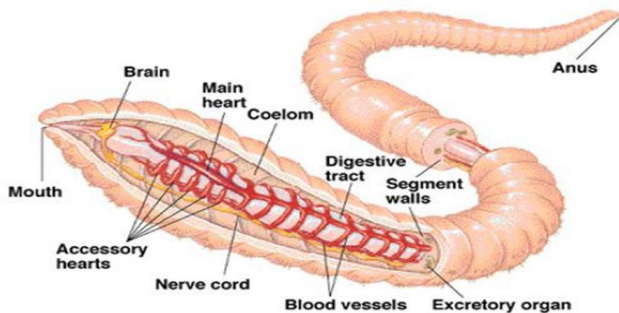


Figure: Showing Digestive system, circulatory system and neural system

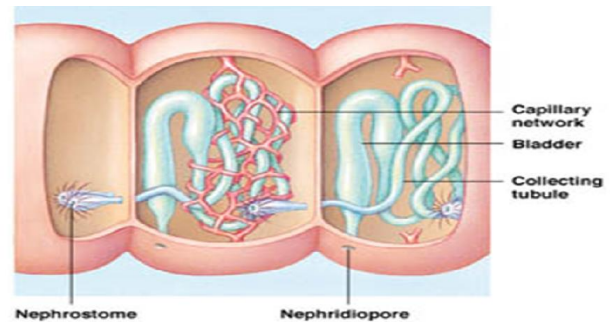
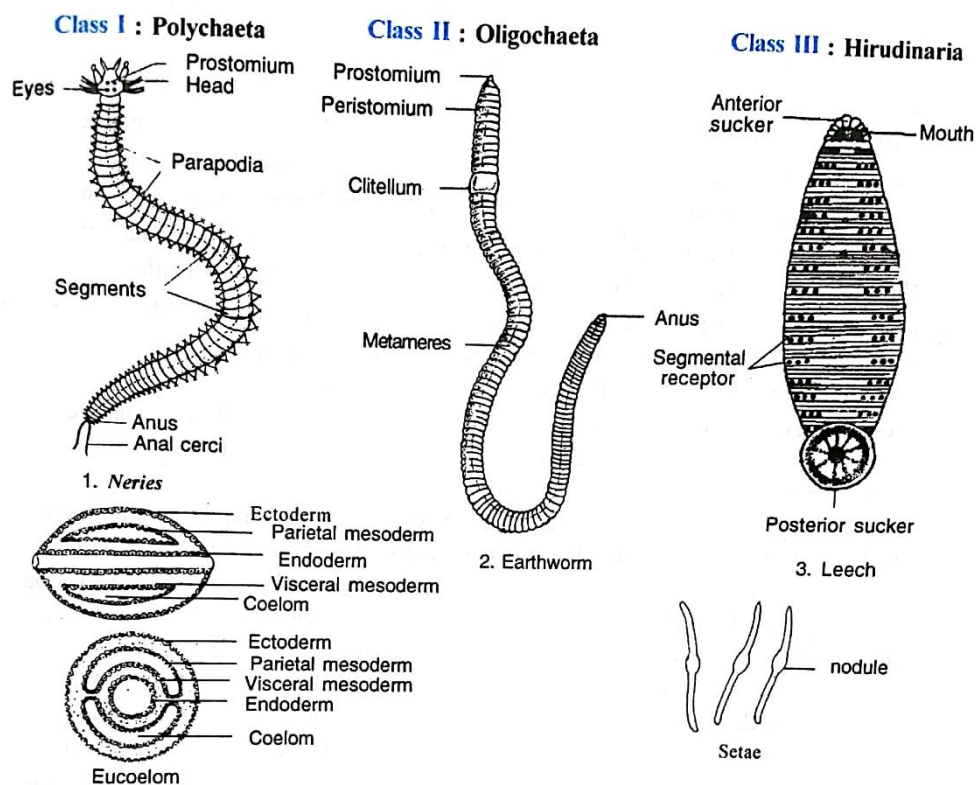


Figure: Excretory system

- Reproduction
 - Strictly sexual
 - Some forms are dioecious (sexes are separate) e.g. *Nereis*. But some are monoecious (hermaphrodites) e.g. **Earthworm** and **Leech**.
- Examples of annelids – *Nereis* (bobbit worm/Clam worm), *Pheretima* (Earthworm), and *Hirudinaria* (leech)



VII. Phylum – Arthropoda (Arthros = joint; poda = appendages)

- This is the largest phylum in Animalia and consists of insects. Over two-third of known species belong to Arthropoda.
- Members have bilateral symmetry.
- These have organ-system level of organisation and are triploblastic and coelomate (**Haemocoel** – coelom filled with blood).
- Body is covered by **chitinous exoskeleton** and has distinct **head, thorax, and abdomen**.
- Have **jointed appendages** (*arthros* – jointed *poda*- appendages)
- Respiratory organs - May have gills/book gills (aquatic)/ book lungs/tracheal system (terrestrial)
- Circulatory system – Open (internal organs are directly bathed by blood)
- Excretory organs - Malpighian tubules.

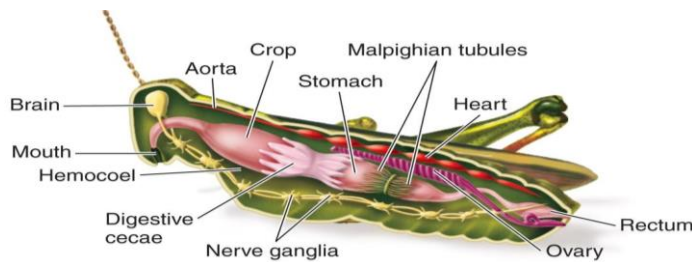


Diagram showing Digestive system, excretory system, nervous system, haemocoel and open circulatory system

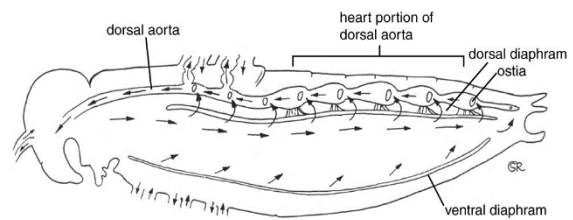


Diagram showing open circulatory system

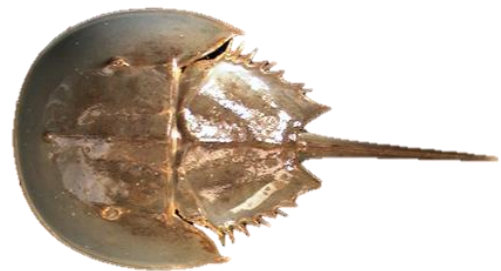
- Sensory organs - Have eyes (compound and simple), antennae, statocysts (balance organs)
- Reproduction
 - Mostly dioecious (sexes are separate)
 - Fertilization is internal and development direct or indirect.
- Examples of arthropods
 - Economically important: - **Apis** (Honey bee), **Bombyx** (silkworm), **Laccifer** (Lac insect)
 - Vectors: - **Aedes** (Mosquito), **Anopheles**, **Culex** – transmits Zeka virus which causes encephalitis in humans and cause avian malaria in birds
 - Pest: - **Locusta** (locust), Cockroach
 - Living fossil – **Limulus** (King crab/Horse shoe crab)



(a) Locust



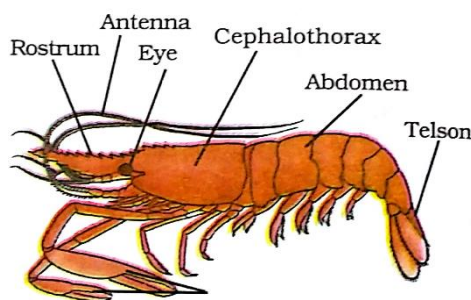
(b) Butterfly



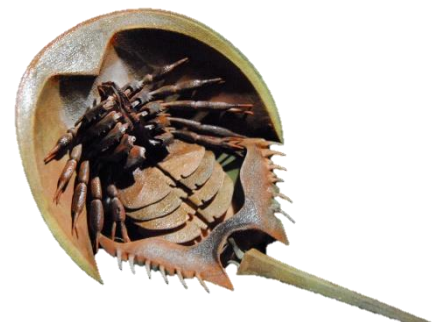
Limulus (dorsal view) (King crab / Horse shoe crab)



(c) Scorpion



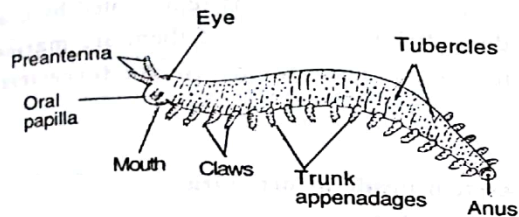
(d) Prawn



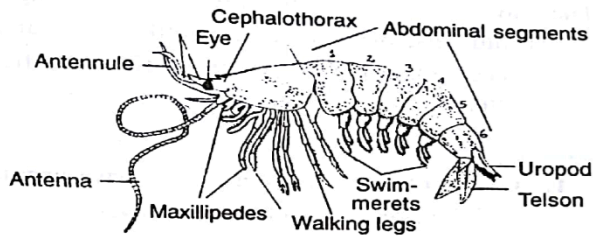
Limulus (ventral view) (King crab / Horse shoe crab)

SYSTEM

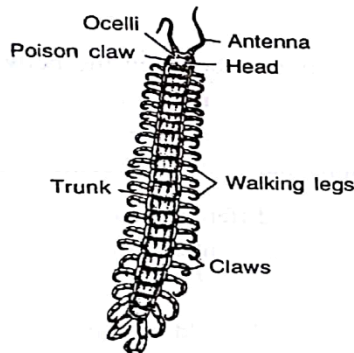
Class I : Onychophora

1. *Peripatus*

Class II : Crustacea

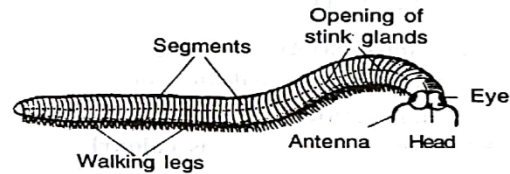
2. *Penaeus*

Class III : Chilopoda



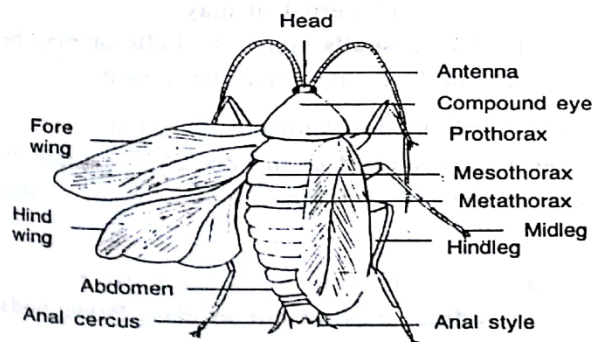
3. Centipede

Class IV : Diplopoda



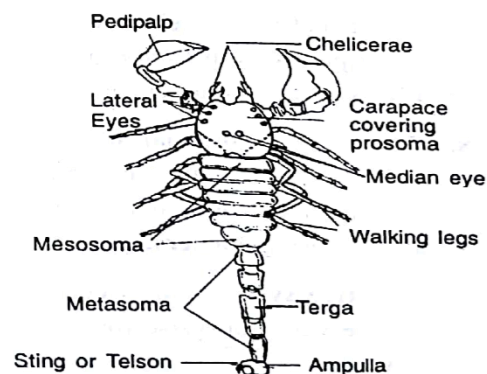
4. Millipede

Class V : Insecta



5. Cockroach

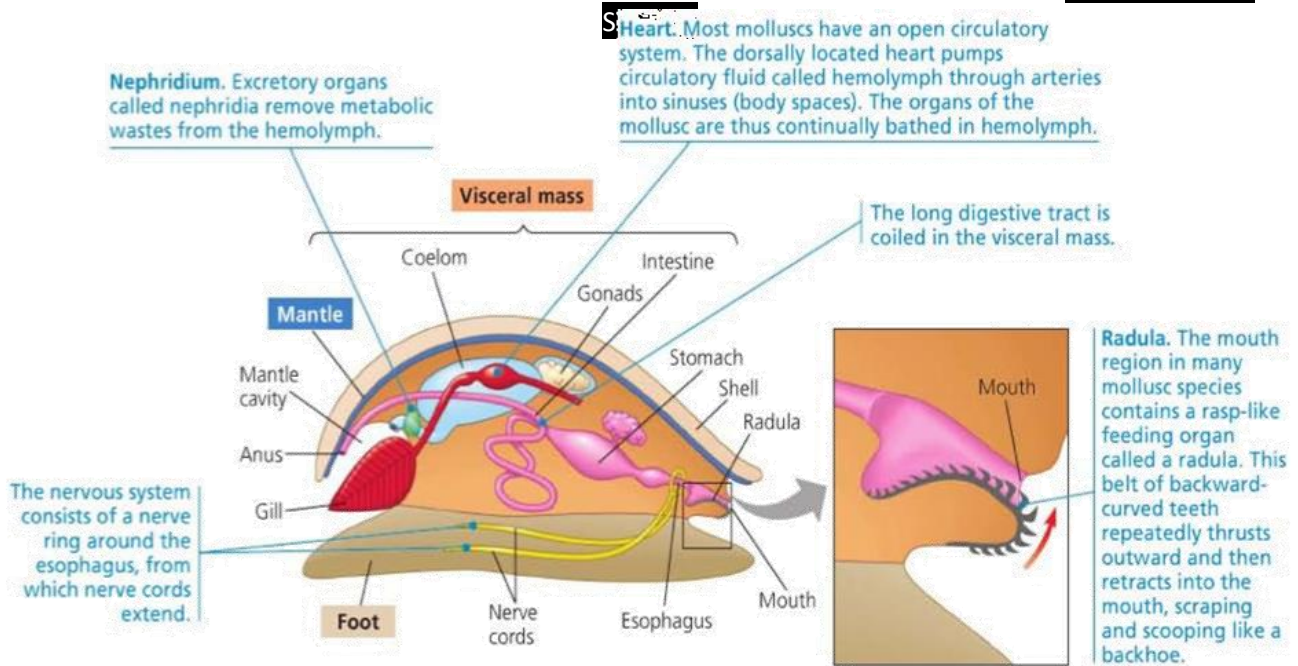
Class VI : Arachnida



6. Scorpion

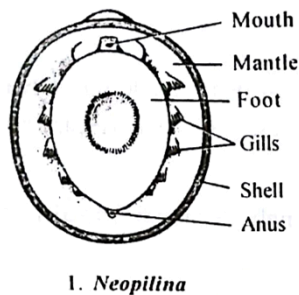
VIII. Phylum – Mollusca (Soft bodied) (L. Mollis = soft)

- **Second largest** phylum in Animalia among invertebrates.
- Habitat: Terrestrial or aquatic (freshwater or marine.)
- Members have bilateral symmetry.
- Have organ system level of organisation; are triploblastic and coelomate.
- Body covered by **calcareous shell**, and has a distinct **head**, a **muscular foot** and a **visceral hump**.
- A soft, spongy layer of skin forms a mantle over the visceral hump.
- Space between the hump and the mantle is known as **mantle cavity**; feather like gills are present in this cavity.
- Gills have both respiratory and excretory functions. Some
- Excretory organs – 1 or 2 pairs of kidneys called **Metanephridia**. Some have organ of Bojanus.
- Sensory organs: Tentacles present on head.
- Feeding organ: **Radula** (file-like.)
- Reproduction:
 - Molluscs are dioecious (sexes are separate) and oviparous (egg laying animals)
 - Development is indirect and direct

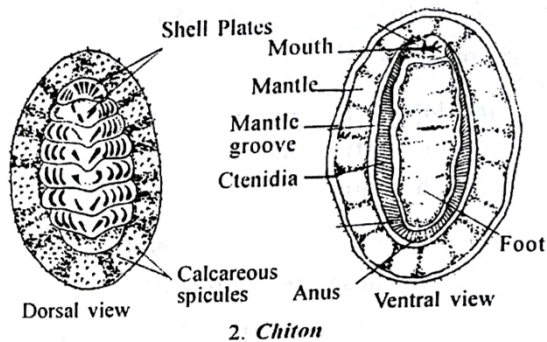


- Examples: ***Pila*** (Apple snail), ***Pinctada*** (Pearl oyster), **Octopus** (Devil fish), ***Sepia*** (Cuttle fish), ***Loligo*** (Squid), ***Aplysia*** (Sea-hare), ***Dentalium*** (Tusk shell) and ***Chaetopleura*** (Chitton.)

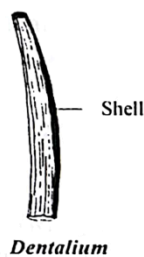
Class I : Monoplacophora



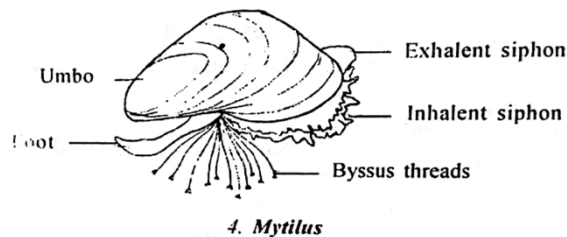
Class II : Polyplacophora



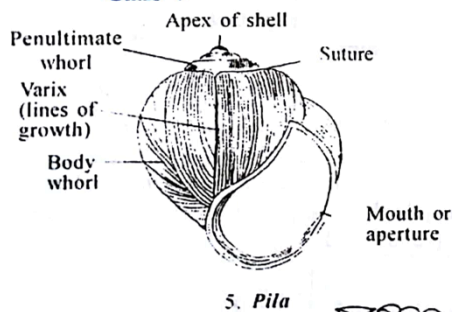
Class III : Scaphopoda



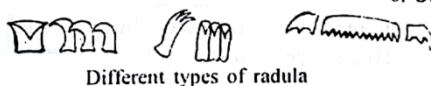
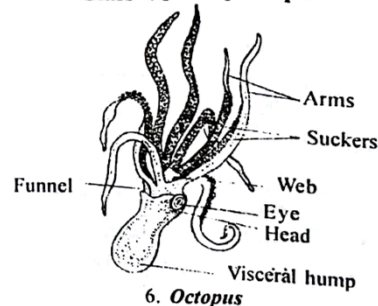
Class IV : Pelecypoda



Class V : Gastropoda



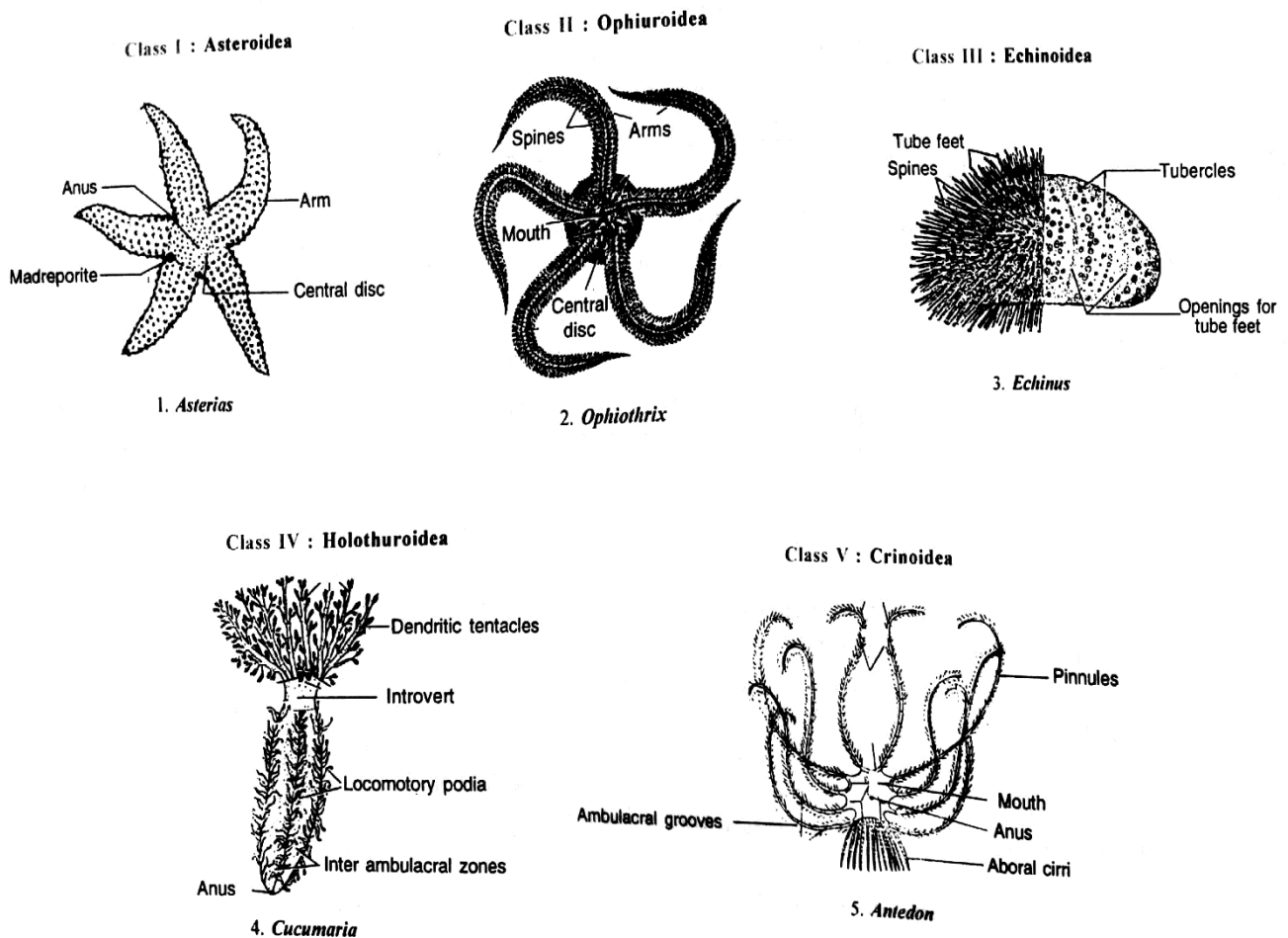
Class VI : Cephalopoda



SYSTEM

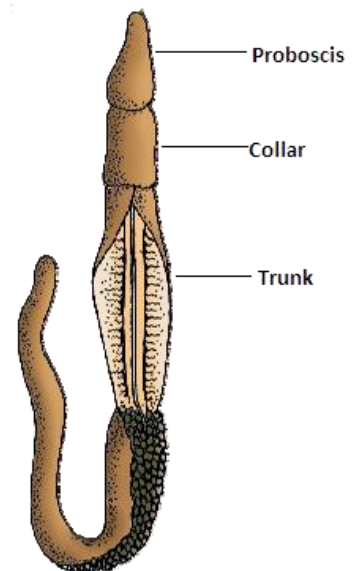
IX. Phylum – Echinodermata (Echinus = spines; derma = skin) (Gr. Echino = hedgehog, derma = skin)

- Echinodermata means spiny-bodied. They are called so because of the presence of an endoskeleton of **calcareous ossicles** (small bones).
- Habitat: Marine.
- Members have bilateral symmetry.
- Have organ system level of organisation; are triploblastic, coelomate.
- Digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side.
- Presence of water vascular system / Ambulacral system:** Distinct feature of this phylum
 - Water vascular system helps in locomotion, capture and transport of food, and in respiration.
- Reproduction:
 - Sexual
 - Fertilisation is external
 - Development is indirect, with the presence of freely swimming larvae stage (Bipinnaria larva)
- Examples: **Asterias** (star fish), **Echinus** (sea urchin), **Cucumaria** (sea cucumber) **Ophiura** (Brittle star), **Antedon** (sea lily)



X. Phylum – Hemichordata (Gr. Hemi = half; Chordos = string)

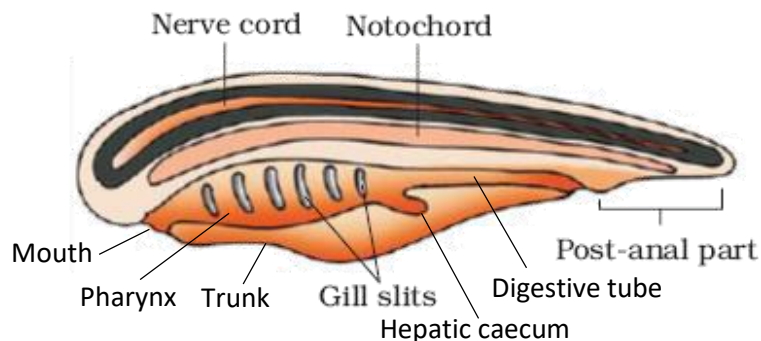
- Earlier placed as a sub-phylum of chordate.
- Consists of a small group of **worm-like** marine animals.
- Members have bilateral symmetry.
- Have organ-system level of organization, are triploblastic and coelomate.
- Body is cylindrical and composed of anterior **proboscis**, **collar**, and a long trunk.
- Circulatory system – **Open**
- Respiratory organs – **Gills**
- Excretory organ – Proboscis gland
- Reproduction:
 - Sexes are separate.
 - Fertilization – external
 - Development – indirect

*Balanoglossus*

- Examples of hemichordates – ***Balanoglossus*** (tongue worm / acorn worm) and ***Saccoglossus***

XI. Phylum Chordata (l. Chorda = rod; ata = group) (Gr. Chordos = string)

- Animals belonging to phylum chordata are fundamental characterized by the presence of
 - **notochord**
 - **dorsal hollow nerve cord**
 - **paired pharyngeal gill slits**



Chordata characteristics

- These are bilateral symmetry, triploblastic organization, and coelomate.
- Organ system level.
- They possess a post anal tail and
- Closed circulatory system.

- Phylum chordata is divided into three **subphyla**:

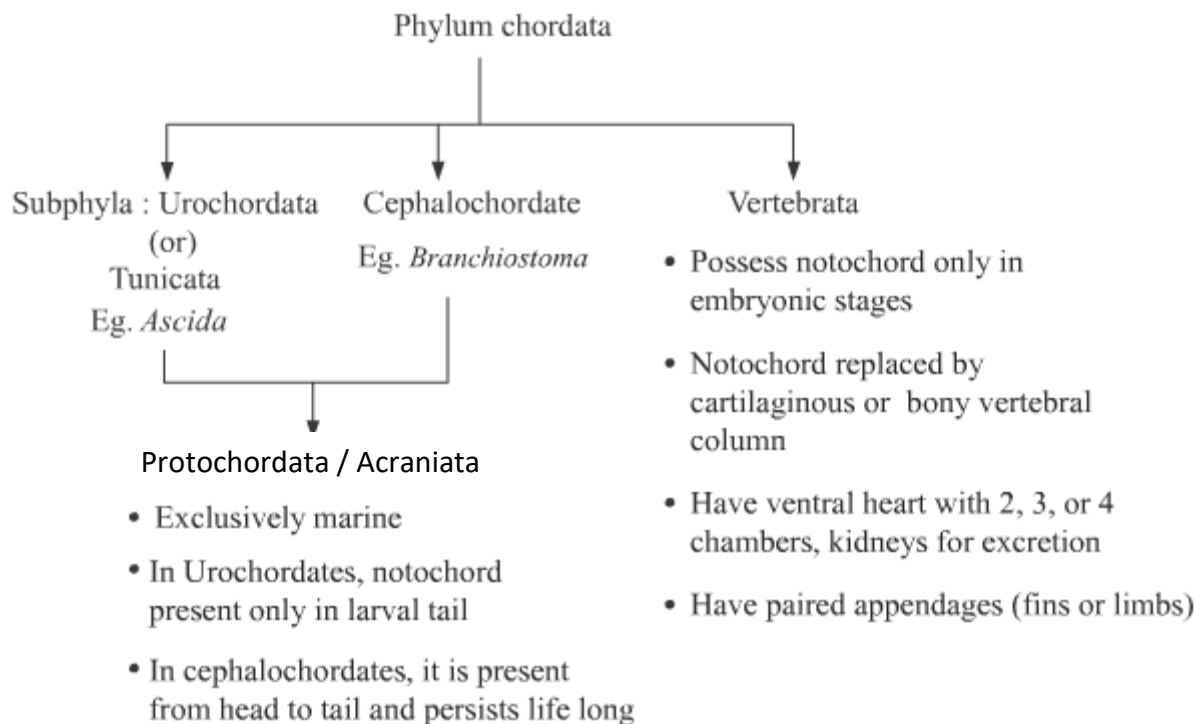
a) Urochordata (Gr. Oura - tail) or **Tunicata** (Gr. Tunic – coat, cover)

E.g. Ascidia, Salpa, Doliolum, Herdmania (sea squirt)

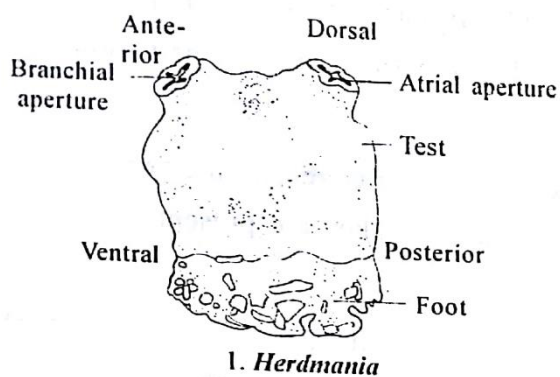
b) Cephalochordates (Gr. Cephalon - head)

E.g. Branchiostoma (Amphioxus or Lancelet)

c) Vertebrata / Craniata*Ascidia*



I. Sub Phylum : Urochordata



II. Sub Phylum : Cephalochordata

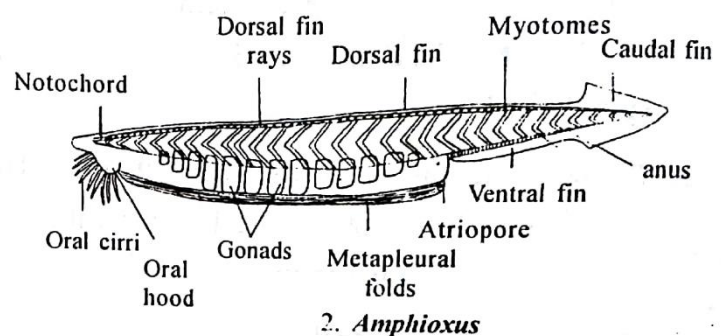
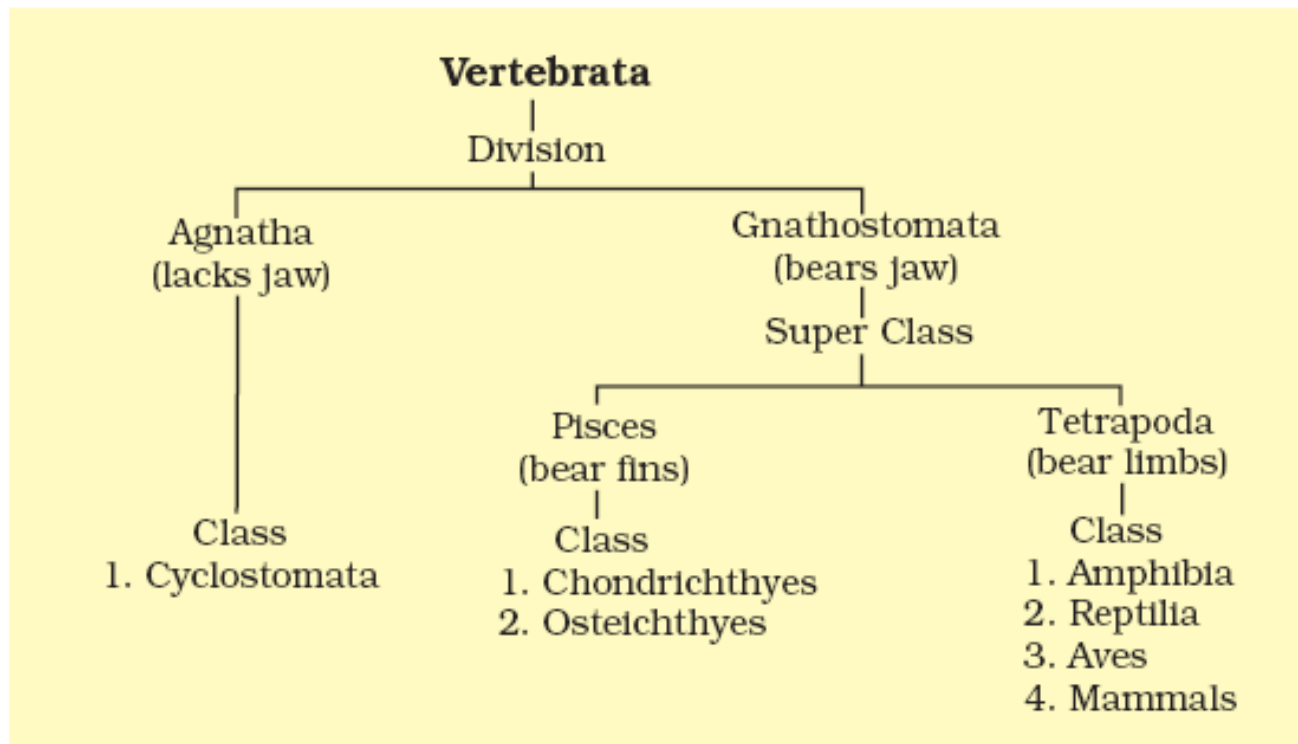


Fig. Protochordates

TABLE - Comparison of Chordates and Non-chordates

S.No.	Chordates	Non-chordates
1.	Notochord present.	Notochord absent.
2.	Central nervous system is dorsal, hollow and single.	Central nervous system is ventral, solid and double.
3.	Pharynx perforated by gill slits.	Gill slits are absent.
4.	Heart is ventral.	Heart is dorsal (if present).
5.	A post-anal part (tail) is present.	Post-anal tail is absent.

- The subphylum Vertebrata is further divided as follows:

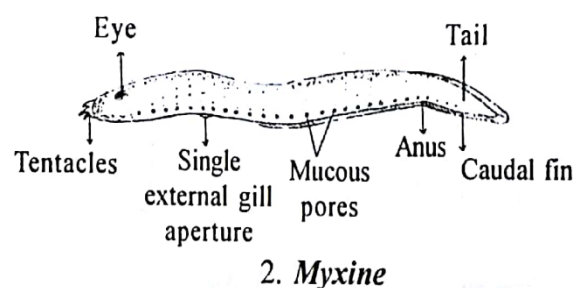
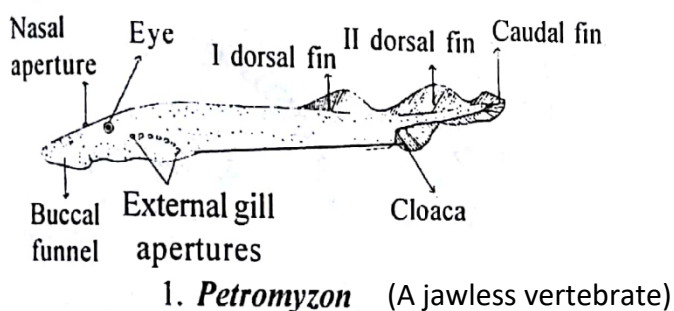


NOTE - "All vertebrates are chordates but all chordates are not vertebrates".

The characteristic features of the phylum, Chordata, include the presence of a notochord and paired pharyngeal gill slits. In sub-phylum Vertebrata, the notochord present in embryos gets replaced by a cartilaginous or bony vertebral column in adults. Thus, it can be said that all vertebrates are chordates but all chordates are not vertebrates.

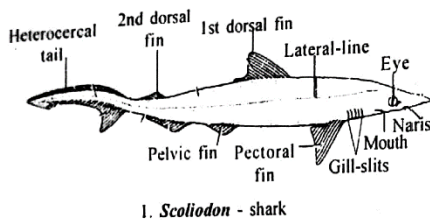
1. Class – Cyclostomata (Gr. Cyclo = circular; Stoma = mouth)

- All members have sucking circular mouth without jaws.
- They have an elongated body (eel like).
- Habitat – they are marine but migrate to fresh waters for spawning. After spawning, within a few days they die and their larvae, after metamorphosis (Development of organism from lower organisation to higher organisation), return to the ocean.
- Nutrition – Ectoparasitic (on other fishes.)
- Respiratory organs – 6-15 pairs of gill slits.
- Body is devoid of scales and paired fins.
- Cranium (the skull) and vertebral column are cartilaginous.
- Examples of cyclostomes – ***Petromyzon*** (Lamprey), ***Myxine*** (Hag fish)

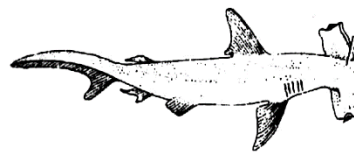


2. Class – Chondrichthyes (Gr. Chondros = cartilage; ichthys = fish)

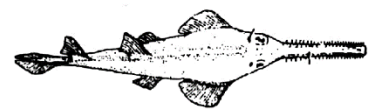
- Habitat – Marine
- Members have **cartilaginous endoskeleton** and streamlined body.
- Notochord is present and persists throughout their life.
- Their skin is tough and contains small **placoid** (plate like shape) **scales**.
- Nutrition – Members of this class are predaceous with strong jaws. The mouth is located ventrally while the teeth, which are actually the modified placoid scales, are backwardly directed.
- Respiratory organs – 5 to 7 pairs of Gills (Gills slits separated without operculum (gill cover).)
- They have to swim continuously to avoid sinking due to the absence of air bladder.
- Heart is two-chambered (one auricle and one ventricle).
- Some have **electric organs** (8 – 220 volts) (Example – **Torpedo**) and some have poisonous sting (Example – **Trygon**-Sting ray.)
- They are cold-blooded animals (**poikilothermous**. i.e., they cannot regulate their body temperature)
- Reproduction:
 - Sexes are separate (dioecious).
 - In males, pelvic fin bears claspers.
 - Fertilization – **Internal**.
 - Many members are **viviparous**.
- Examples of Chondrichthyes – **Scoliodon** (dog fish), **Pristis** (Saw fish), **Carcharodon** (Great white shark), **Trygon** (Sting ray).



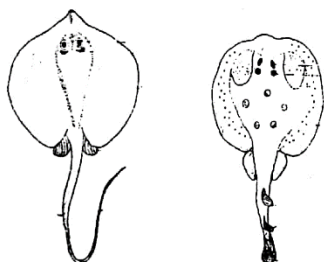
1. *Scoliodon* - shark



2. *Zygaena sphyryna* - hammer headed fish

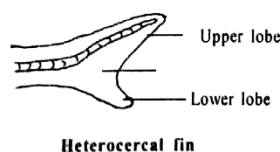


3. *Pristis* - saw fish

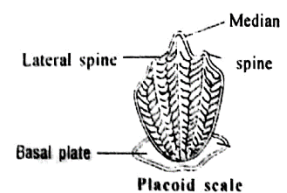


4. *Trygon* - Sting ray

5. *Narcine* - Electric ray



Heterocercal fin

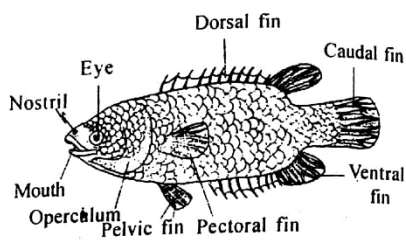
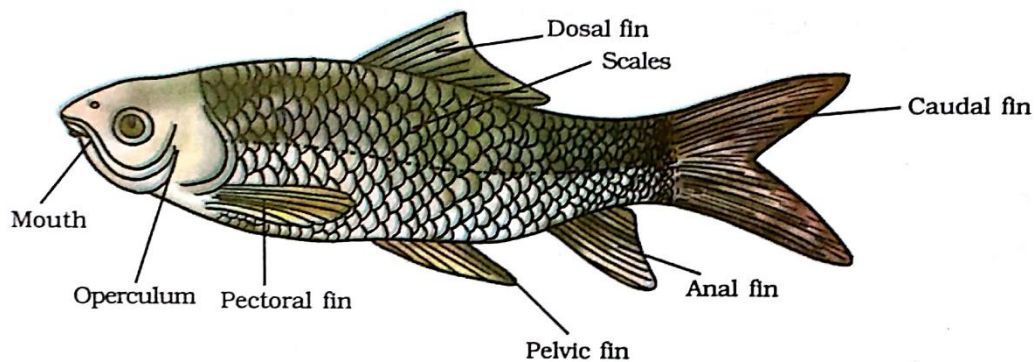
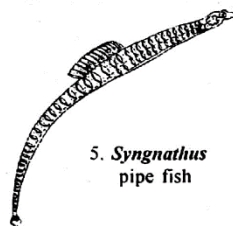


Placoid scale

3. Class – Osteichthyes (Gr. Osteos = bone; ichthys = fish)

- Habitat – both marine and freshwater.
- Members have **bony endoskeleton** and streamlined body with mouth terminally placed.
- Body covered by cycloid (a kind of curve) or ctenoid scales.
- Respiratory organ – Four pairs of gills covered by operculum; presence of **air bladder** provides buoyancy.
- Heart is two-chambered (one auricle and one ventricle)
- They are also poikilothermous.
- Reproduction:
 - Sexes are separate.

- Fertilization – **External**.
- Mostly oviparous.
- Development – **Direct**.
- Example of Osteichthyes -
 - Marine – **Exocoetus** (Flying fish) **Hippocampus** (Sea horse),
 - Freshwater – **Labeo** (Rohu), **Catla** (Katla), **Clarias** (Magur)
 - Aquarium – **Betta** (Fighting fish), **Pterophyllum** (Angel fish)

1. *Anabas* - climbing perch2. *Anguilla* - eel3. *Exocoetus* - flying fish4. *Hippocampus* - sea horse5. *Syngnathus*
pipe fish

Homocercal fin



Diphyccercal fin



Ctenoid scale



Cycloid scale

*Betta* (fighting fish)*Pterophyllum* (Angel fish)

SYSTEM

Difference between Chondrichthyes and Osteichthyes	
Chondrichthyes	Osteichthyes
• Skeleton is made up of cartilage	• Skeleton is made up of bones.
• Exclusively marine	• Both marine and fresh water forms are seen.
• They possess placoid scales.	• Possess cycloid or ctenoid scales
• Mouth is situated on ventral side of the body.	• Mouth is terminal in position
• Possess 5 pairs of gillslits.	• Possess 4 pairs of gills slits which is covered by operculum
• Quadal tail is heterocercal.	• Qadual tail is homocercal.
• Devoid of air bladders.	• Possess air bladders
• Males usually possess a pair of claspers	• Claspers are absent in males
• Examples – Scoliodion (shark), Trygon (sting ray)	• Examples – hippocampus (sea horse), clarias (Magur / cat fish)

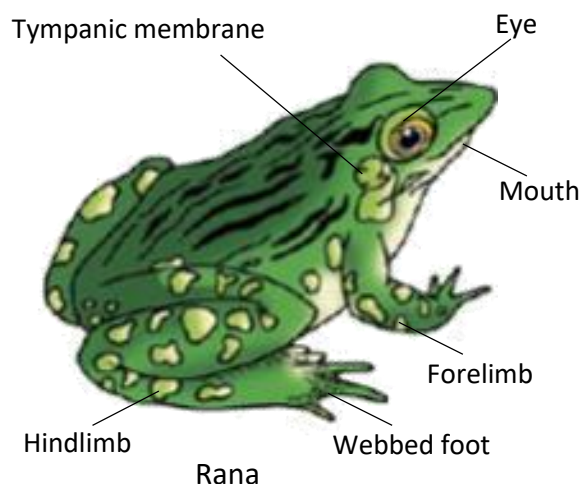
4. Class – Amphibia (Gr. Amphi = double; bios = life)

- Can live both in water and on land.
- Body divisible into **head** and **trunk** with two pairs of limbs; tail may be present in some.
- Alimentary canal, excretory and reproductive tracts open into common opening called **cloaca** on the exterior.
- Skin is moist without scales.
- Eyes have eyelids and ear is represented by **tympanum**.
- Respiration can be by gills, lungs, and skin.
- Heart is three-chambered (two auricles and one ventricle).
- Poikilothermous (cold blooded animals)
- Reproduction:
 - Sexes are separate.
 - Fertilization – External.
 - They are oviparous.
 - Development – Indirect.

Examples of amphibians – **Bufo** (toad), **Rana** (Frog), **Salamandra** (Salamander), **Hyla** (Tree frog), **Ichthyopsis** (Limbless amphibian)



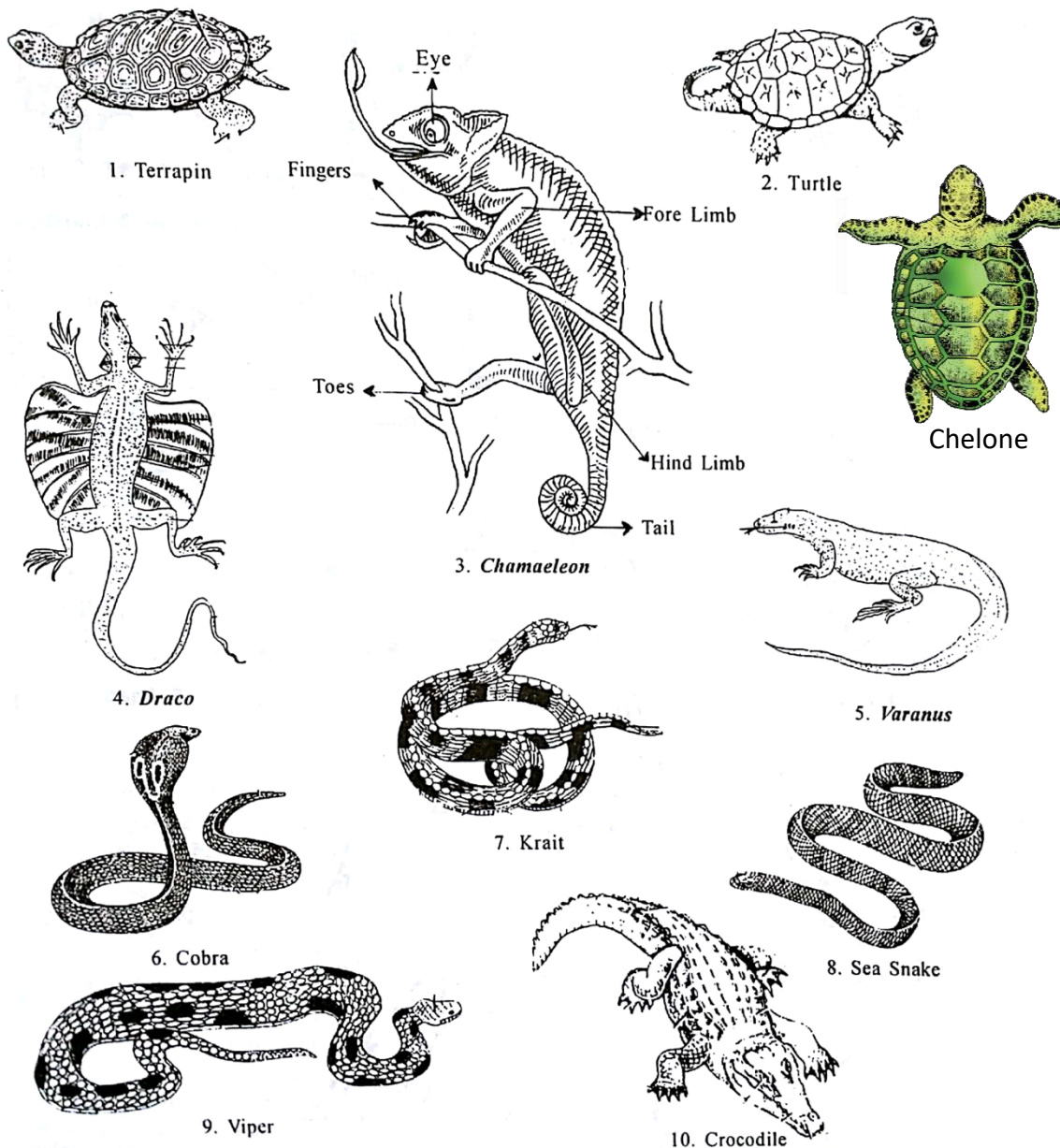
Salamander



Rana

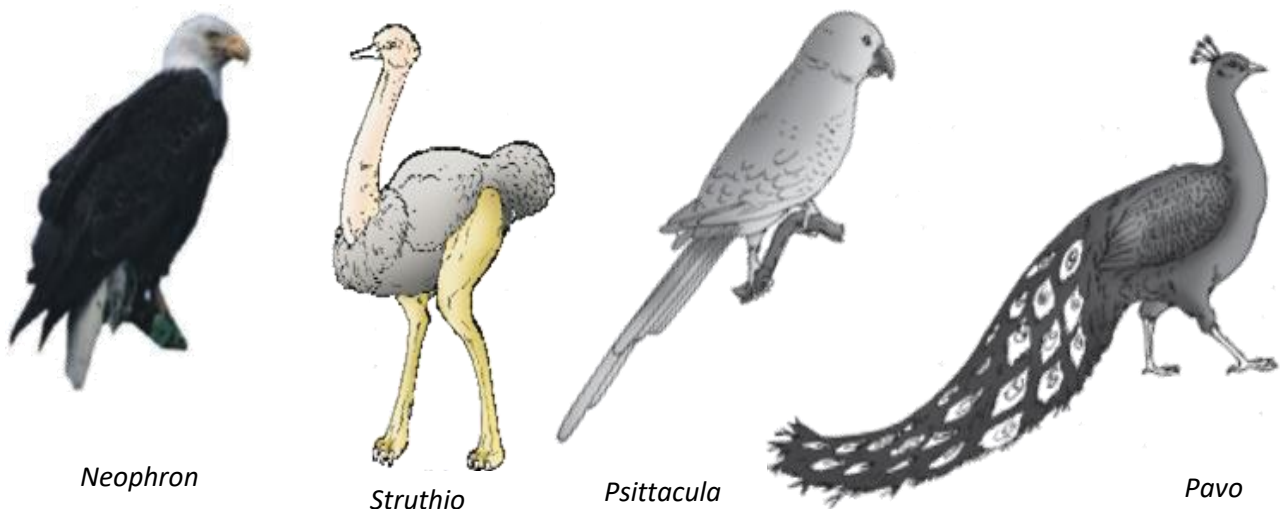
5. Class – Reptilia (L. Repere or reptum = to creep or to crawl)

- Characterized by creeping, crawling mode of locomotion.
- Habitat – Mostly terrestrial.
- Body covered by dry cornified skin, epidermal scales of scales or scutes.
- Do not have external ears, tympanum present.
- Two pairs of limbs, when present.
- Heart is three-chambered, but it is four-chambered in crocodiles.
- Poikilothermous.
- Snakes and lizards shed their scales as skin cast.
- Reproduction:
 - Sexes are separate.
 - Fertilization is internal. Reptiles are **oviparous**.
 - Development – Direct.
- Examples - *Chelone* (Turtle), *Testudo* (Tortoise), *Chamaeleon* (Tree lizard), *Calotes* (Garden lizard), *Crocodilus* (Crocodile), *Alligator* (Alligator). *Hemidactylus* (Wall lizard), Poisonous snakes – *Naja* (Cobra), *Bangarus* (Krait), *Vipera* (Viper).



6. Class – Aves (L. Avis = Birds)

- Characterized by the presence of wings and ability to fly with exception of flightless birds (e.g. Ostrich)
- They have beaks.
- Forelimbs modified into wings and hind limbs have scales; used for walking, swimming, or clasping tree branches.
- Skin is dry with oil glands present only at the base of the tail.
- Skeletal system – Endoskeleton is ossified (bony) and long bones are hollow with **air cavities** (pneumatic.)
- Respiratory organ – Lungs and air sacs.
- Digestive system – have two additional chambers: crop and gizzard.
- Heart is four-chambered.
- They are **homoiothermous** (warm-blooded.) i.e. they are able to maintain a constant temperature.
- Reproduction:
 - Fertilization – Internal; are oviparous.
 - Development – Direct.
 - Sexes are separate.
- Example – **Corvus** (Crow), **Columba** (Pigeon), **Psittacula** (Parrot), **Struthio** (Ostrich), **Pavo** (Peacock), **Aptenodytes** (Penguin), **Neophron** (Vulture)



7. Class – Mammalia (L. Mamma = breast)

- Habitat – Variety of habitats – polar ice, deserts, mountains, forest, grassland and dark caves.
- Show adaptations that enable them to fly, to swim, or to walk.
- Presence of **mammary glands** (produce milk for young ones) – unique characteristic of mammals.
- Respiratory organ – Lungs.
- Have two pairs of limbs adapted for walking, running, climbing, burrowing, swimming or flying.
- Skin is covered by hair (Present only in mammals.)
- External ears (**Pinnae**) are present.
- Heart is four-chambered. These are homoiothermous.
- Reproduction:
 - Sexes are separate.
 - Fertilization – Internal.
 - They are viviparous with exceptions (Example – Platypus which is oviparous.)
 - Development – Direct.

SYSTEM

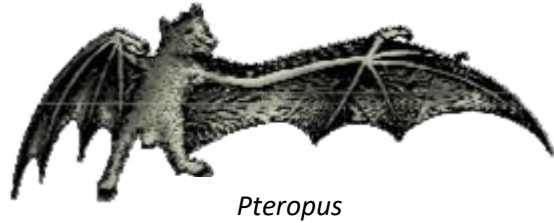
- Examples of mammals – Oviparous-*Ornithorhynchus* (Platypus); Viviparous - *Macropus* (Kangaroo), *Pteropus* (Flying fox), *Camelus* (Camel), *Macaca* (Monkey), *Rattus* (Rat), *Canis* (Dog), *Felis* (Cat), *Elephas* (Elephant), *Equus* (Horse), *Delphinus* (Common dolphin), *Balaenoptera* (Blue whale), *Panthera tigris* (Tiger), *Panthera leo* (Lion)



Ornithorhynchus



Macropus



Pteropus



Balaenoptera



1. Platypus



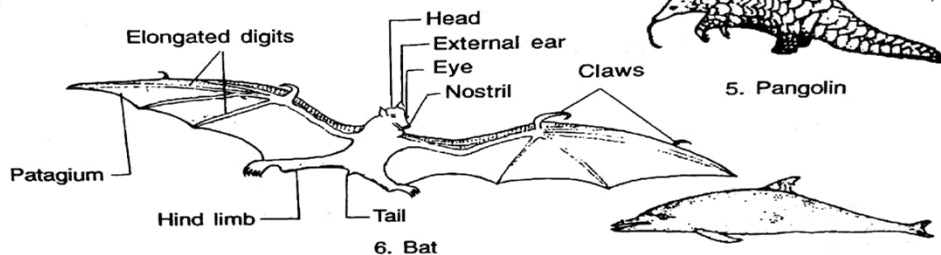
2. Echidna



3. Kangaroo



4. Koala



6. Bat



5. Pangolin



7. Blue whale



9. Mouse



12. Horse



13. Zebra



14. Giraffe



10. Squirrel



11. Rabbit



15. Camel



16. Deer



18. Elephant



17. Rhinoceros



19. Wolf



20. Cat



22. Monkey



21. Lion

Broad classification of Kingdom Animalia based on common fundamental features

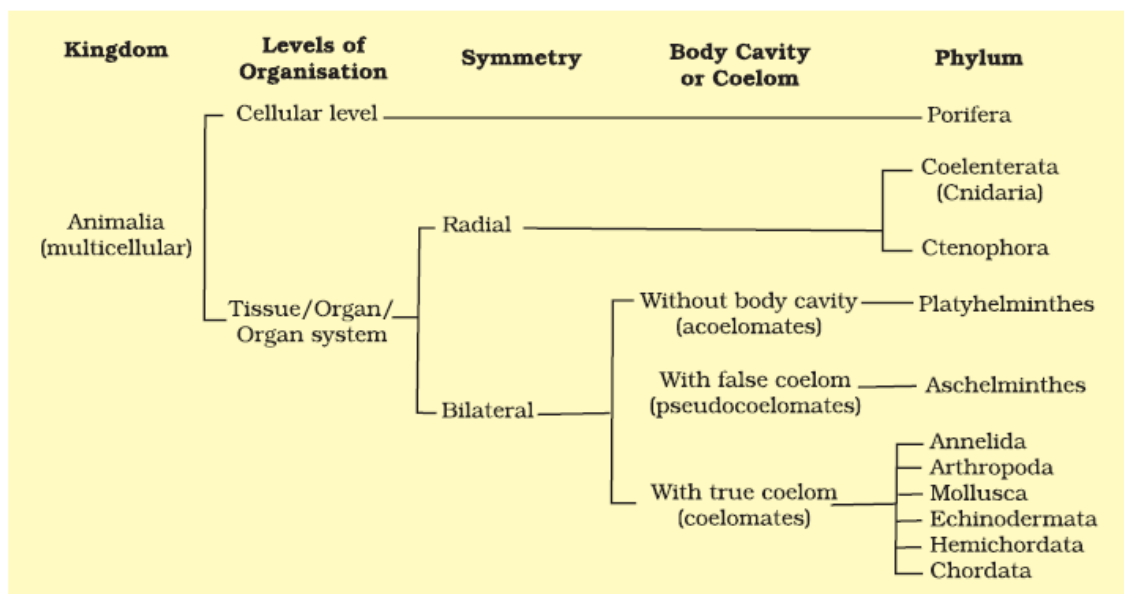


TABLE : Salient Features of Different Phyla in the Animal Kingdom

Phylum	Level of Organisation	Symmetry	Coelom	Segmentation	Digestive System	Circulatory System	Respiratory System	Distinctive Features
Porifera	Cellular	Many	Absent	Absent	Absent	Absent	Absent	Body with pores and canals in walls.
Coelenterata (Cnidaria)	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Cnidoblasts present.
Ctenophora	Tissue	Radial	Absent	Absent	Incomplete	Absent	Absent	Comb plates for locomotion.
Platyhelminthes	Organ & Organ-system	Bilateral	Absent	Absent	Incomplete	Absent	Absent	Flat body, suckers.
Aschelminthes	Organ-system	Bilateral	Pseudo coelomate	Absent	Complete	Absent	Absent	Often worm-shaped, elongated.
Annelida	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Body segmentation like rings.
Arthropoda	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Exoskeleton of cuticle, jointed appendages.
Mollusca	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	External skeleton shell usually present.
Echinodermata	Organ-system	Radial	Coelomate	Absent	Complete	Present	Present	Water vascular system, radial symmetry.
Hemichordata	Organ-system	Bilateral	Coelomate	Absent	Complete	Present	Present	Worm-like with proboscis, collar and trunk.
Chordata	Organ-system	Bilateral	Coelomate	Present	Complete	Present	Present	Notochord, dorsal hollow nerve cord, gill slits with limbs or fins.

SUMMARY

The basic fundamental features such as level of organisation, symmetry, cell organisation, coelom, segmentation, notochord, etc., have enabled us to broadly classify the animal kingdom. Besides the fundamental features, there are many other distinctive characters which are specific for each phyla or class.

Porifera includes multicellular animals which exhibit cellular level of organisation and have characteristic flagellated choanocytes. The coelenterates have tentacles and bear cnidoblasts. They are mostly aquatic, sessile or free-floating. The ctenophores are marine animals with comb plates. The platyhelminthes have flat body and exhibit bilateral symmetry. The parasitic forms show distinct suckers and hooks. Aschelminthes are pseudocoelomates and include parasitic as well as non-parasitic round worms.

Annelids are metamerically segmented animals with a true coelom. The arthropods are the most abundant group of animals characterised by the presence of jointed appendages. The molluscs have a soft body surrounded by an external calcareous shell. The body is covered with external skeleton made of chitin. The echinoderms possess a spiny skin. Their most distinctive feature is the presence of water vascular system. The hemichordates are a small group of worm-like marine animals. They have a cylindrical body with proboscis, collar and trunk.

Phylum Chordata includes animals which possess a notochord either throughout or during early embryonic life. Other common features observed in the chordates are the dorsal, hollow nerve cord and paired pharyngeal gill slits. Some of the vertebrates do not possess jaws (Agnatha) whereas most of them possess jaws (Gnathostomata). Agnatha is represented by the class, Cyclostomata. They are the most primitive chordates and are ectoparasites on fishes. Gnathostomata has two super classes, Pisces and Tetrapoda. Classes Chondrichthyes and Osteichthyes bear fins for locomotion and are grouped under Pisces. The Chondrichthyes are fishes with cartilaginous endoskeleton and are marine. Classes, Amphibia, Reptilia, Aves and Mammalia have two pairs of limbs and are thus grouped under Tetrapoda. The amphibians have adapted to live both on land and water. Reptiles are characterised by the presence of dry and cornified skin. Limbs are absent in snakes. Fishes, amphibians and reptiles are poikilothermous (coldblooded). Aves are warm blooded animals with feathers on their bodies and forelimbs modified into wings for flying. Hind limbs are adapted for walking, swimming, perching or claspings. The unique features of mammals are the presence of mammary glands and hairs on the skin. They commonly exhibit viviparity.